It is well-known that the digit sum (in base 10) preserves the remainder modulo 9.

This is easy to prove: all powers of 10 give the remainder 1 when divided by 9, and therefore $\sum a_i \cdot 10^i$ (the value of the number whose digits are a_i) gives the same remainder as $\sum a_i$ (the digit sum of that number).

This gives us a **necessary condition** for the solvability of our task: whenever two numbers x and y share the same digit sum, they must share the same remainder modulo 9, and therefore their difference d must be divisible by 9. In other words, if d isn't a multiple of 9, the answer is NONE.

The public subtasks are both solvable by brute force: print NONE if d isn't divisible by 9, and in all other cases iterate over all pairs $(x, x+d)$ until you find one where the digit sum matches. It turns out that for each solvable d you'll find such a pair pretty quickly.

For the full solution we need to come up with a more clever construction. There are many possibilities how to do it, but one of them is definitely both simplest and most clever.

As d is a multiple of 9, let $d = 9k$. Now, consider the numbers k and $10k$. Their difference is clearly $9k = d$. And their digit sums are clearly the same, as $10k$ is just formed from k by appending a zero.